MODULE 4 OF 5

Injury-Specific Treatment Pathways

~38 min read Running Gait Analysis for Injury Prevention & Treatment

4.1 From Hypothesis to Treatment Plan

Module 3 gave you a set of gait-based hypotheses for each of the five dominant running injuries. This module takes each pathology and builds the treatment pathway that follows once a working diagnosis is confirmed on history, palpation, and special testing — not gait findings alone.

A consistent thread runs through all five conditions below: strengthening and progressive loading of the relevant muscle groups have the strongest and most consistent evidence base of any intervention category. Gait retraining is addressed separately in 4.7 and built out fully in Module 5 — it is a targeted tool for specific presentations, not a default first-line intervention for every runner who walks through the door.

4.2 Patellofemoral Pain

Given the hip-adduction mechanism identified in Module 3, it follows that hip- and knee-targeted strengthening is the best-supported intervention for PFP, and the trial evidence bears this out directly.

Intervention	Evidence & Notes
Combined hip and knee strengthening Exercises: Bridge · Alternating leg lifts in bridge · Squat · Lunge with band perturbation · Y-balance reach · Balance on foam pad · Step-down · Prone hip internal rotation (banded) · Standing hip internal rotation (banded)	A randomised trial comparing a combined hip-and-knee strengthening protocol against knee-strengthening alone found the combined group achieved significantly better pain and function outcomes, supporting a hip-inclusive approach over a knee-only quadriceps programme.
Hip abductor and external rotator strengthening Exercises: Side-lying hip abduction · Clamshell · Fire hydrant · Prone hip external rotation (banded) · Banded lateral walk · Banded wall slide · Standing hip abduction (banded) · Standing hip external rotation (banded) · Seated hip external rotation (banded)	Targets the same hip adduction pattern flagged as the strongest prospective predictor in Module 3; typically progressed from open-chain (side-lying, banded) through to closed-chain, functional loading.
Quadriceps strengthening Exercises: Quadriceps setting (isometric) · Terminal knee extension (banded) · Straight-leg raise (VMO emphasis) · Wall slide with ball squeeze · Squat to target · Squat · McConnell squat · McConnell squat – with ball · Bulgarian split squat · Lunge · Step-up · Step-down	Remains a core component; PFP has long been associated with quadriceps weakness independent of the hip mechanism, and closed-chain strengthening is generally preferred to isolated open-chain knee extension for symptom tolerance.
Load management	Reducing aggravating volume or intensity in the short term while strength is built, rather than complete rest, which does not address the underlying capacity deficit.

Source: Nascimento LR, Teixeira-Salmela LF, Souza RB, Resende RA. Hip and Knee Strengthening Is More Effective Than Knee Strengthening Alone for Reducing Pain and Improving Activity in Individuals With Patellofemoral Pain: A Systematic Review With Meta-analysis. J Orthop Sports Phys Ther. 2018;48(1):19–31.

4.3 Iliotibial Band Syndrome

ITBS treatment follows directly from the reduced hip abductor strength identified consistently in the Module 3 literature, and a systematic review of hip-strengthening interventions specifically for ITBS supports the approach.

Intervention	Evidence & Notes
Hip abductor strengthening Exercises: Side-lying hip abduction · Standing hip abduction (banded) · Clamshell · Fire hydrant · Banded lateral walk · Banded wall slide	A systematic review found hip abductor strengthening programmes were associated with reductions in ITBS pain reported across included studies ranging from roughly 27% to 100%, alongside improved hip abductor strength.
Gluteus medius & posterior chain strengthening Exercises: Bridge · Alternating leg lifts in bridge · Single-leg bridge · Bridge – feet on chair · Bridge – feet on ball · Bridge with ball roll-in · Lateral step-up · Lunge with band perturbation · Y-balance reach · Step-down	Typically progressed from isolated abduction work to functional single-leg loading (step-downs, single-leg bridges) that better replicates the demands of stance phase.
Short-term load reduction	Reducing volume, hill running, or cambered-surface running while symptomatic and strength is being rebuilt.
Foam rolling / soft tissue work Exercises: Foam rolling – lateral thigh	Commonly used adjunctively for symptom modulation; evidence supports short-term symptomatic relief rather than addressing the underlying strength deficit, so it should not replace strengthening.

Source: Ferber R, Kendall KD, Pohl MB. Runners with iliotibial band syndrome demonstrate atypical hip and knee kinematics. J Orthop Sports Phys Ther. 2010;40(2):52–58; Fredericson M, Wolf C. Iliotibial band syndrome in runners: innovations in treatment. Sports Med. 2005;35(5):451–459.

4.4 Medial Tibial Stress Syndrome

BEFORE TREATING

Re-confirm the red-flag screening from Module 2 (2.7) before beginning a loading-based programme. Focal bony tenderness, night pain, or pain at rest should prompt referral for imaging to rule out tibial bone stress injury before proceeding with the approach below.

MTSS OR BONE STRESS INJURY?

Typical of MTSS

TENDERNESS
Diffuse, along the posteromedial border of the distal two-thirds of the tibia, usually over 5 cm or more

PAIN WITH RUNNING
Often present early in a run and may ease as the runner warms up

AT REST
Settles with rest; no night pain

HOPPING
Usually tolerated

NEXT STEP
Graded loading, as below

Red flags for bone stress injury

TENDERNESS
Focal, pinpoint bony tenderness

PAIN WITH RUNNING
Builds through the run and can persist afterwards

AT REST
Pain at rest, at night, or with walking

HOPPING
Painful single-leg hop

NEXT STEP
Stop running and refer for imaging

Typical presentations only; findings overlap, so when in doubt, treat as a bone stress injury until imaging rules it out. Tenderness length criterion: Yates B, White S. The incidence and risk factors in the development of medial tibial stress syndrome among naval recruits. Am J Sports Med. 2004;32(3):772–780.

Once bone stress injury has been screened out, MTSS management centres on progressive, graded loading of the lower leg musculature rather than passive modalities alone.

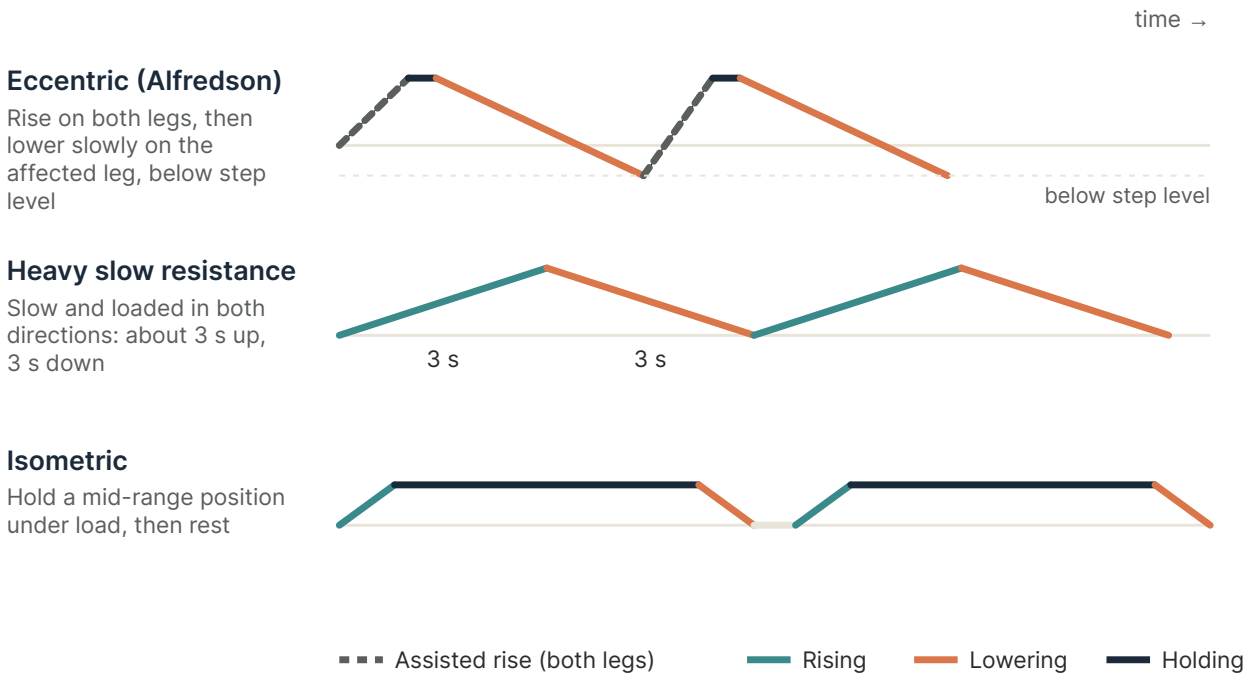
Intervention	Evidence & Notes
Progressive calf and tibialis posterior loading Exercises: Calf raise progression	A structured lower-leg exercise programme, progressed from isometric through to plyometric loading over weeks, has been shown to improve outcomes compared to standard care in runners with MTSS.
Ankle dorsiflexion mobility work Exercises: Knee-to-wall dorsiflexion	Targets the reduced static dorsiflexion range identified as a consistent finding in Module 3; a stiff ankle may drive the compensatory eversion pattern implicated in this condition.
Graded return-to-run volume	MTSS is fundamentally a training-load condition, so any structural intervention needs to sit alongside deliberate volume and intensity management, covered fully in Module 5.

Source: Moen MH, Holtslag L, Bakker E, et al. The treatment of medial tibial stress syndrome in athletes; a randomized clinical trial. Sports Med Arthrosc Rehabil Ther Technol. 2012;4:12; Becker J, James S, Wayner R, Osternig L, Chou LS. Am J Sports Med. 2017;45(11):2614–2621.

4.5 Achilles Tendinopathy

Loading is the cornerstone of evidence-based Achilles tendinopathy management. The two best-studied protocols are eccentric-only loading and heavy slow resistance (HSR), and head-to-head trial evidence has clarified how they compare.

Intervention	Evidence & Notes
Eccentric loading (Alfredson protocol) Exercises: Eccentric heel drop (Alfredson)	The original and most widely studied protocol; strong evidence for mid-portion Achilles tendinopathy specifically, less consistent results for insertional presentations.
Heavy slow resistance (HSR) Exercises: Heavy slow resistance calf raise	A randomised trial comparing HSR directly against eccentric training found both produced comparable improvements in pain and function at 12-month follow-up, with runners reporting higher satisfaction and better adherence to the HSR protocol.
Isometric loading Exercises: Isometric calf hold	Commonly used in the acute, highly irritable stage for short-term analgesic effect before progressing into one of the isotonic protocols above.



Schematic heel position over two repetitions for each protocol. The protocols differ mainly in which direction is loaded and how fast, not just in the exercise itself.

Given the Module 3 finding on forefoot strike increasing Achilles load, a runner who has self-adopted a forefoot pattern to manage a different (e.g. patellofemoral) complaint and now presents with Achilles symptoms is a specific scenario where re-examining that gait choice — not just prescribing more loading — belongs in the plan.

Source: Beyer R, Kongsgaard M, Hougs Kjær B, et al. Heavy Slow Resistance Versus Eccentric Training as Treatment for Achilles Tendinopathy: A Randomized Controlled Trial. Am J Sports Med. 2015;43(7):1704–1711.

4.6 Plantar Fasciitis

Plantar fasciitis treatment has shifted over the past two decades from stretching-only protocols toward an approach that combines fascia-specific stretching with progressive strength training.

Intervention	Evidence & Notes
Plantar-fascia-specific stretching	A randomised trial found plantar-fascia-specific stretching produced significantly better outcomes than standard Achilles-stretching protocols at 8 weeks, and the benefit was maintained at long-term follow-up.
High-load strength training Exercises: High-load heel raise with toe extension · Calf raise progression · Big toe flexion (banded) · Big toe flexion in standing (banded)	More recent trial evidence supports progressive strength training of the plantar flexors (calf raises, progressing to loaded single-leg variants) as comparable or superior to stretching alone for medium-term outcomes, with the added benefit of addressing broader lower-limb loading capacity.
Footwear and load review	A practical check on cushioning, wear, and recent changes in training surface or volume, alongside the strength-and-stretch programme rather than as a substitute for it.

Source: DiGiovanni BF, Nawoczenski DA, Lintal ME, et al. Tissue-specific plantar fascia-stretching exercise enhances outcomes in patients with chronic heel pain. J Bone Joint Surg Am. 2003;85(7):1270–1277; Rathleff MS, Mølgaard CM, Fredberg U, et al. High-load strength training improves outcome in patients with plantar fasciitis: A randomized controlled trial. Scand J Med Sci Sports. 2015;25(3):e292–e300.

4.7 Chronic Exertional Compartment Syndrome

Where CECS has been confirmed (typically with compartment pressure testing after conservative measures have failed, or strong clinical suspicion while that referral is pending), gait modification has trial-level support as a genuine alternative to fasciotomy — worth raising explicitly with runners who associate CECS only with surgery.

Intervention	Evidence & Notes
Forefoot-strike retraining	A six-week programme has been shown to reduce post-running intracompartmental pressure and maintain pain and disability improvements for up to a year, avoiding surgical intervention entirely across the study's patients.
Gradual exposure	As with the fault-pattern retraining in 4.8, change is introduced incrementally (stationary drills before running, short intervals before full sessions) rather than switching a runner's strike pattern abruptly.
Monitor symptom response directly	Using the runner's own distance-to-onset marker as a simple, patient-specific outcome measure as the programme progresses.

Source: Diebal AR, Gregory R, Alitz C, Gerber JP. Forefoot Running Improves Pain and Disability Associated With Chronic Exertional Compartment Syndrome. Am J Sports Med. 2012;40(5):1060–1067.

4.8 Treating the Three Gait Fault Patterns

The three fault patterns introduced in Module 3 (3.8) each have a fairly consistent retraining approach in the literature, built on the same principles as the cadence retraining in Module 5: external cueing, gradual exposure, and fading feedback over roughly 2 weeks / 8 sessions rather than a single corrective instruction.

Pattern	Retraining Approach
Over-striding	Metronome or music cueing to the target cadence; technological feedback devices where available; standing drills that rehearse a forward "free-falling" posture before adding running; built up gradually over roughly 8 sessions across 2 weeks, with feedback faded as running time increases.
Cross-over	Mirror or real-time video feedback; verbal cues; a visual target (road markings, or tape on a treadmill belt) to run either side of; deliberately decreasing stride length; gradually increasing running time while decreasing feedback, following the same fade pattern as over-striding.
Medial collapse	Mirror or real-time video feedback with markers placed over the specific points of concern (e.g. pelvis, knee); progressing from stationary, to walking, to running; decreasing stride length; gradually increasing running time while decreasing feedback.



Retraining set-ups. Left: mirror in front of the treadmill, with a tape line down the belt as a visual target for crossover. Centre: skin markers over the points of concern (PSIS, greater trochanter, mid thigh, lateral knee joint, mid tibia, lateral malleolus). Right: a metronome app for cadence cueing.

Note the common structure across all three: visual or auditory feedback that is deliberately withdrawn over time, built on a foundation of the condition-specific strengthening already covered in 4.2–4.6 — gait retraining is layered onto that strength work, not used instead of it.

4.9 When Gait Retraining Belongs in the Plan

Every condition above is treated first and foremost with progressive loading and strengthening — that is where the strongest evidence sits, and it is where most treatment plans should start. Gait retraining is not a replacement for that foundation; it is an additional tool for a specific subset of runners.

> **CLINICAL REASONING**
>
> Consider gait retraining as an adjunct when a runner has a clearly identified, modifiable gait pattern from Module 2's assessment (most commonly overstriding / low cadence, given the strength of the cadence evidence covered in Module 1), symptoms that have plateaued despite an adequate strengthening programme, or a pattern directly implicated in their specific injury from the Module 3 mapping.

WHY STRENGTH TRAINING ALONE DOESN'T FIX GAIT

It's tempting to assume that once a runner is strong enough, their gait fault will simply resolve on its own — the evidence doesn't support that assumption. Strengthening and skill training appear to activate different areas of the motor cortex, and skill training produces cortical reorganisation in a way that strengthening alone does not. Hip strengthening and movement retraining, when not run specifically in the context of running itself, have been shown not to alter abnormal running mechanics. The practical implication: a runner needs *running-specific* neuromuscular retraining — not just generic gym-based strengthening — to actually change a gait fault, even once they're strong enough to support the change.

Sources: Jensen JL, Marstrand PCD, Nielsen JB. Motor skill training and strength training are associated with different plastic changes in the central nervous system. J Appl Physiol. 2005;99(4):1558–1568. Willy RW, Davis IS. The effect of a hip-strengthening program on mechanics during running and during a single-leg squat. J Orthop Sports Phys Ther. 2011;41(9):625–632.

Module 5 builds out exactly how to run a gait-retraining intervention — cueing strategies, dosage, and how to integrate it with the load-management and return-to-run planning that ties this whole course together.

4.10 Example Progressive Programme

The tables above set out *what* to load for each condition; this section shows one way of putting it together over time. It combines the two arms of a multicentre randomised trial in patellofemoral pain — a six-week hip- and core-strengthening protocol and a six-week knee-focused protocol — into a single programme that loads both joints. Each week uses five exercises, balanced between hip- and knee-biased work, at three sessions per week, so total volume stays close to either original protocol rather than doubling it. Every exercise links to its card in the Exercise Library below.

Week	Exercise	Focus	Sets	Reps / time
Week 1 Isolated, low-load activation	Clamshell	Hip	3	10
	Side-lying hip abduction	Hip	3	10
	Bridge	Hip	3	10
	Straight-leg raise (VMO emphasis)	Knee	3	10
	Squat to target (quarter depth)	Knee	3	10
Week 2 Added resistance, early closed chain	Clamshell (banded)	Hip	3	15
	Prone hip external rotation (banded)	Hip	3	10
	Alternating leg lifts in bridge	Hip	3	10 each side
	Wall slide with ball squeeze (quarter depth)	Knee	3	30 s hold
	Squat to target (half depth)	Knee	3	15

Week	Exercise	Focus	Sets	Reps / time
Week 3 Stronger bands, partial-range loading	Banded lateral walk	Hip	3	10 steps each way
	Prone hip internal + external rotation (stronger band)	Hip	3	10 each
	Single-leg bridge	Hip	3	10 each side
	Squat (half depth)	Knee	3	10
	Banded wall slide (quarter depth)	Knee + hip	3	10
Week 4 Single-leg and step work	Fire hydrant	Hip	3	10 each side
	Bridge – feet on ball	Hip	3	30 s hold
	Lateral step-up	Knee + hip	3	10 each side
	Step-down (low step)	Knee + hip	3	10 each side
	Lunge (quarter depth)	Knee	3	10 each side

Week	Exercise	Focus	Sets	Reps / time
Week 5–6 Functional, balance and full range to 90°	Bridge with ball roll-in	Hip	3	10–15
	Y-balance reach	Hip + knee	3	5 per direction each side
	Lunge with band perturbation	Knee + hip	3	30 s each side
	Step-down (higher step)	Knee + hip	3	15 each side
	Lunge to max 90° knee flexion	Knee	3	15 each side

PATIENT HANDOUT

A printable version of this programme for your patients: plain-language instructions with a photo of each exercise, the 3/10 pain rule, and tick boxes for every session (3 pages, A4). Download it from section 4.10 of the Module 4 lesson page.

How the programme progresses

- **Weeks 1–2** — isolated, low-load work to build hip abductor and external rotator capacity and quadriceps activation, mirroring the opening week of both source protocols.
- **Weeks 3–4** — stronger bands and partial-range closed-chain loading (half squats, wall slides, step-ups and step-downs from a low step), matching the point where both source protocols increase band resistance and introduce single-leg work.
- **Weeks 5–6** — single-leg, functional and balance tasks to a maximum of 90° knee flexion. Y-balance reach and perturbation lunges take the place of the source protocol's foam-pad balance work.

- **Depth and step height** progress from quarter to half to 90° knee flexion, following the knee-focused protocol.
- **Frequency** — three sessions per week throughout.

> **PAIN-MONITORING RULE**
>
> Progress to the next stage when the current exercises are completed with pain no higher than **3/10** during exercise and no increase in symptoms the following morning. If pain rises above 3/10 during a session, or symptoms are worse the next day, hold at the current stage or step back one stage before progressing again.

> **CLINICAL NOTE**
>
> This is one worked example of translating the evidence into a programme, not a fixed prescription. The same structure suits ITBS, where the hip-abductor emphasis maps directly onto 4.3. Exercise selection should be adjusted to the runner's specific deficits, irritability and available equipment, and return-to-run progression runs alongside it — covered in Module 5.

Source: Ferber R, Bolgla L, Earl-Richardson JE, Pohl MB, Willy RW. Strengthening of the hip and core versus knee muscles for the treatment of patellofemoral pain: a multicenter randomized controlled trial. J Athl Train. 2015;50(4):366–377.

4.11 Exercise Library

The exercises referenced in the treatment tables above, gathered in one place and grouped by region. Each card shows the main target, a short execution cue, and which conditions it is used for. Dosage and progression should follow the condition-specific evidence in each section rather than a fixed prescription.

> **EXERCISE VIDEOS ONLINE**
>
> The exercises are listed here by name only. A looping video and execution cue for every exercise is in the online Exercise Library: Module 4 lesson page, section 4.11, at gait.corelearning.co.za.

HIP & PELVIS

- Bridge
- Alternating leg lifts in bridge
- Single-leg bridge
- Bridge – feet on chair
- Bridge – feet on ball
- Bridge with ball roll-in
- Side-lying hip abduction
- Clamshell
- Fire hydrant
- Prone hip external rotation (banded)
- Prone hip internal rotation (banded)
- Banded lateral walk
- Banded wall slide
- Standing hip abduction (banded)
- Standing hip external rotation (banded)
- Seated hip external rotation (banded)
- Seated hip internal rotation (banded)
- Standing hip internal rotation (banded)
- Foam rolling – lateral thigh

KNEE & CLOSED-CHAIN CONTROL

- Quadriceps setting (isometric)
- Terminal knee extension (banded)
- Terminal knee extension – foot on ball
- Banded knee extension (seated)
- Straight-leg raise (VMO emphasis)
- Wall slide with ball squeeze
- Squat to target
- Squat
- McConnell squat
- McConnell squat – with ball
- Bulgarian split squat
- Lunge
- Lunge with band perturbation
- Step-up
- Lateral step-up
- Balance on foam pad
- Y-balance reach
- Step-down

LOWER LEG & FOOT

- Calf raise progression (6 stages)
- Knee-to-wall dorsiflexion
- Eccentric heel drop (Alfredson)
- Heavy slow resistance calf raise
- Isometric calf hold
- High-load heel raise with toe extension
- Big toe flexion (banded)
- Big toe flexion in standing (banded)

Summary

Progressive, condition-specific strengthening is the common thread across all five injuries: combined hip-and-knee work for PFP, hip abductor strengthening for ITBS, graded lower-leg loading for MTSS (after red-flag screening), eccentric or heavy-slow-resistance loading for Achilles tendinopathy, and fascia-specific stretching paired with strength training for plantar fasciitis. Gait retraining sits alongside this foundation as a targeted adjunct for runners with a clearly identified, modifiable pattern — not a first-line substitute for loading. Module 5 now turns to how gait retraining and structured return-to-run programming are actually delivered.